

Dysgu: A Tool to Keep Students Engaged Outside the Classroom^{*}

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Abstract— This paper presents an out-of-class active learning environment, called Dysgu. Dysgu presents an innovative approach to out-of-class activities by combining multiple dimensions of best practices from different spectrum of student learning into a coherent idea and delivering such activities with personalization and adaptation. The goal of the Dysgu system is to study the impact of frequent out-of-class activities on student learning and engagement when the students can compare their progress with the rest of the class and where the activities are smaller (in scope) with scaffolding support, are interactive in nature, and delivered via a mobile platform. Initial usability tests and software engineering quality matrices show that the software is easy to use, manage and extend.

Keywords—Active learning, mobile learning, homework, out-of-class activity.

I. INTRODUCTION

It is fairly accepted that [1]-[2] freshman students are not as prepared for college work as their predecessors. The lack of preparation is exacerbated by the lack of practice outside the classroom. College advising guideline [3] states that students should study a specific number of hours at home per credit hour of class per week. However, it is found [2] that students seldom follow such guidelines. A possible reason [3] for the lack of study outside the class is that students are working increasing hours on a job outside the classroom. Reasons contributing to the rise in working hour include rising tuition cost, rising living expenses, and other socio-cultural challenges. Working more hours inevitably leads to students spending less time studying outside the class, which have detrimental effects on student's learning. Therefore, it is critical to developing methods that will ensure effective usage of the time that the students have for studying outside the classroom in such socio-economic situations.

To address this challenge, this paper presents an active learning environment, Dysgu, that enables teachers to create interactive out-of-class activities that are delivered in a structured way to facilitate greater student engagement, better practice of learned concepts, and improved student learning outside the classroom. This is achieved by providing students with personalized and active learning activities that they can do during their most convenient times using a widely-used platform (i.e., mobile), thus utilizing their time and maximizing their engagement with the course content.

II. RATIONAL FOR MOBILE OUT-OF-CLASS ACTIVITIES

Homework or out-of-class activities are an integral part of a traditional face-to-face classroom. However, in recent years, such activities have been scrutinized [4] because of the

lack of consideration for the individual learner's learning needs, different learning styles, personal situations, student's aspirations, class context, and motivation. Although online and distance learning attempts to provide such personalization, the lack of interaction between faculty and students in a purely online environment makes it difficult to keep the students engaged outside the classroom. Therefore, a more blended approach where face-to-face teaching is supplemented with personalized and interactive activities outside the classroom has the potential to support students learning and engagement outside the classroom. Since mobile learning environments can deliver content anytime and anywhere, by having a mobile-based active learning environment, which adapts to student's needs, can assist faculty in guiding student's knowledge acquisition; well after students leave the classroom.

III. DYSGU

Dysgu is designed as a cloud-based client-server system as shown in Fig. 1. The client and server do not have any direct communication between them. All interactions between the server and client in Dysgu is delegated through cloud services. The major differences of Dysgu with a similar system are the mode of interaction, type of activities, ways it provides personalization and ability to adapt to students' needs. More on that will be discussed in section IV.

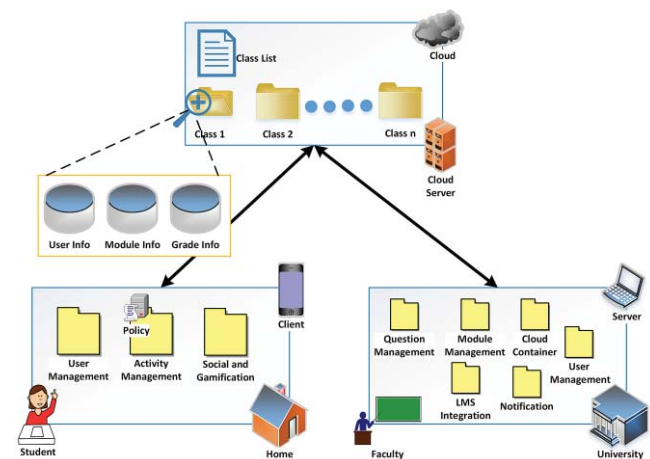


Fig. 1 Software architecture of Dysgu.

In Dysgu's software architecture, the cloud stores information about different classes. There is a collection of repositories for each class, which is maintained in the cloud. These repositories can be updated by both server or clients and changes are propagated through asynchronous access. The cloud services are completely hidden from both the faculty and the students and managed transparently by

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Dysgu. The cloud repository is used as an intermediary for the client and the server to communicate with each other in complete synchronicity and transparency. Dysgu utilizes free services offered by the cloud providers to make the system free to use.

The faculty uses the server (Fig. 2) to manage the system, which includes creating and deploying a class, managing students and their credentials, creating interactive exercises, creating and deploying problem-solving modules, creating different reports, monitoring student progress and class status, etc. A client application (Fig. 3) in student device allows students to log in to the class, work on a problem-solving session, monitor progress etc. Both the client and the server is developed to run asynchronously without each other. To facilitate that, the cloud repositories are used in the background to provide transparent and seamless operation.

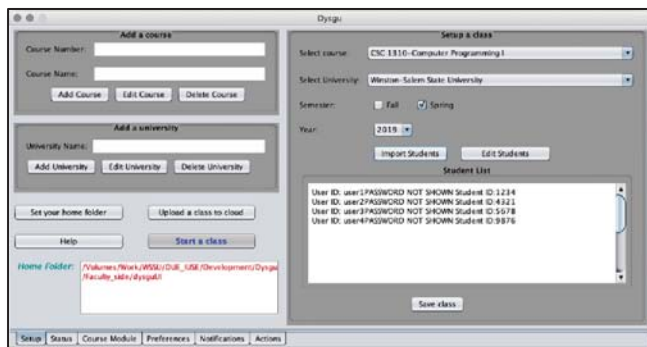


Fig. 2 Server interface.

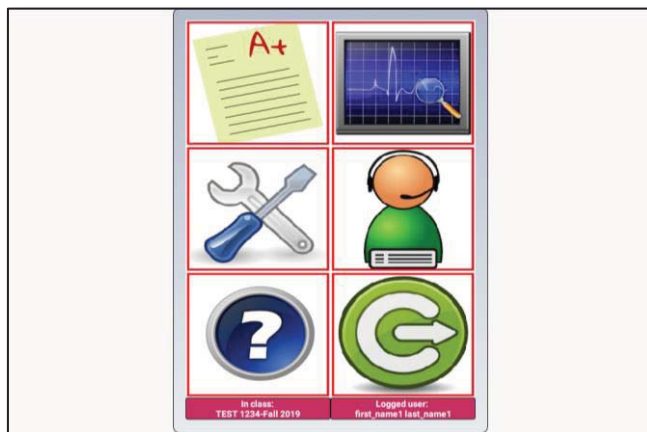


Fig. 3 Client interface.

Because Dysgu is designed to facilitate interactive problem solving, it should be used in classes which is skill generating. This allows the faculty to utilize different cognitivist and constructivist principles to enhance the student's skills. Dysgu comes with a set of interactive activities (which are highly parameterized), however, anyone can build and incorporate new activities in Dysgu using published API. Although Dysgu can be used with any level of college students, we envision it to be used with freshman and sophomore students. Students at this level usually struggle with factors related to transitioning from high school to college, and Dysgu can positively affect their learning and engagement outside the classroom as older students are usually matured enough to handle such factors.

A. Interactive Exercise

Dysgu supports interactive problem solving for out-of-class activities. Interactive problems [5] are a visual representation of a problem, where students can interact with different steps or parts of it by selecting or touching different components of the visualization. Most such interactive problem-solving session is comprised of multiple interactive screens and students are allowed to navigate those screens. This allows students to see the effect of their interaction in real-time. The properties of an interactive activity in Dysgu are as follow:

- There must be interactive elements (such as buttons, lists, tables, dropdown boxes, etc.) where students have to physically interact with those elements (by dragging, dropping, rearranging, etc.) to come up with the answer. Fig. 4, shows one such example where students are given a list of elements, which they have to classify into different beans. Students can drag and drop any of the items to any of the beans or rearrange them as they like. Some of the items can also have images and will zoom open when students touch them or started dragging them.

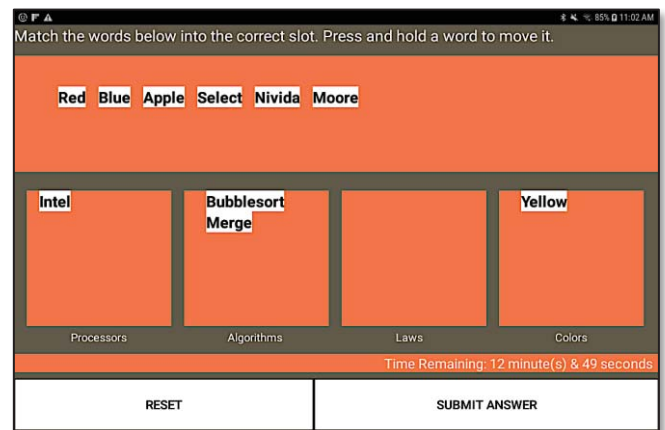


Fig. 4 Single screen interactive activity

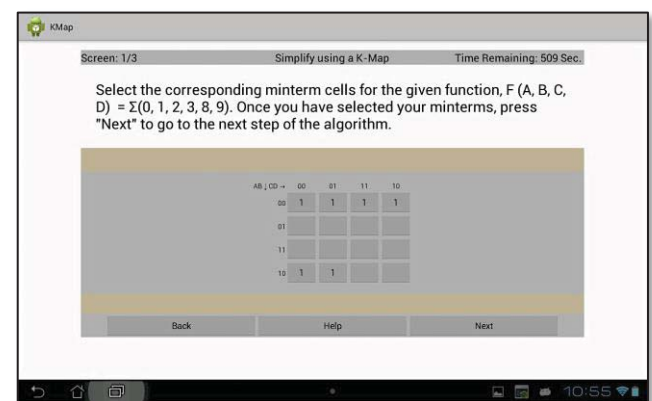


Fig. 5 Multiple screen interactive activity.

- There can be multiple screens in the problem-solving activity, each of the screens having the same property as listed in (a) above. Additionally, each screen should have a way to traverse to the next (or the previous) screen, which allows the students to see the effects of their interaction in one screen to another. Fig. 5 shows one such interactive problem-solving activity, where students have to traverse through multiple screens to

solve the problem. At any time, students can traverse any direction, change their choices, see effects of their choices and modify their choices if the traversing caused them to change their mind.

Any interactive activity is not tied to any single problem or exercise. It can be populated with any number of similar problems by feeding it with the new problem. Therefore, faculty can utilize one such activity to test students with different problems with a different degree of difficulties. Since the client grades the student's answer and provides a set of feedback, faculty workload is not increased. These interactive activities can be developed by anyone and incorporated with Dysgu following published APIs. Additionally, to adapt to student's asynchronous lifestyle and schedule, such activities in Dysgu can be paused and resumed before the due date corresponding to the learning path and course module's policy. This allows the students to work on a problem in their own time and pace.

B. Learning Paths

Course modules in Dysgu is a collection of interactive problem-solving activities, organized into a list of learning paths, where each activity in a path have certain relationships. Normally, problems in a learning path have a different degree of difficulties which gradually increase from easy to hard for examining student's comprehension of certain concepts. Fig. 6 shows a typical Dysgu module. General properties of a module are as follows:

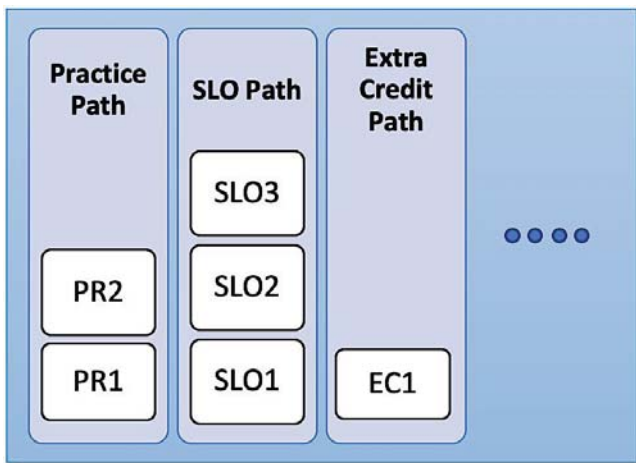


Fig. 6 Course modules in Dysgu

- A module can have a total of five paths, each path can have five activities in it. The reason for these sizes is set considering student workload, allowable time and learning outcomes.
- Some of the path or paths in a module can be designated as a practice path (s) to help students practice the concept with different problems. Therefore, such a practice path is not graded, however, students can see whether their answer is correct or not and can earn points.
- Other paths can be designated as student learning outcome (SLO) paths, where students will be assessed of their learning skills. Therefore, activities in these path(s) have scores and no points (Section III.C) and the score is used to grade the student as part of the course.
- Additionally, faculty can designate some paths as extra credit, which have problems that require additional effort

from students. These paths are also not graded, but students might be assigned points depending on their performance. This allows students the freedom to try-and-fail without penalizing their course grades. Research [6] has found that, in such instances, students actually do better in solving harder problems and utilize higher order thinking.

C. Scoring and comparisons

Dysgu uses two grading schemes: scores, and points. A score can be assigned to any activity in a SLO path, which can only be positive and the total of all scores in SLO path (s) should add up to 100 (to make calculations easier). If a student successfully solves an activity, a corresponding score will be assigned to that student, which is later used to calculate the student's overall grade in the class. Points (can be negative) on the other hand are assigned as reward of finishing an activity in the system. For instance, finishing an activity first in class or finishing an activity in one try will give students positive points; whereas unfinished activities will produce negative points. Scores are utilized to calculate student's grade, whereas, points are used as currency in the system. Students can use accumulated points as currency to extend the time to solve a module or buy hints/tries to solve harder problems. Students can also obtain badges depending on the amount and type of points. In order to enhance student motivation and competency, Dysgu provides information related to the peers of the class, such as the number of peers already completed an activity, the number of peers progressing toward stage t of a path, etc. Moreover, Dysgu presents a glimpse of progress for a student in comparison to the rest of the class. It shows the student score, where the student stands compared to the class, how many points the student has and what are the badges the student earned. For each activity in a path, the students can see how much they have completed, how much the class has completed, how many points the activity has and how long they have to complete the activity. Students can review completed activities and compare their performance with the rest of the class. Dysgu provides a summative information about the class without disclosing individual student grades to follow federal privacy laws.

D. User Experience

Dysgu uses a workflow-based approach to give faculty an intuitive step-oriented process to use the software. A workflow usually comprises of a sequence of steps through which the faculty is guided by the software. Fig. 7 shows some of the major workflows in the server-side of the Dysgu system. For example, an initial workflow is to create a class. To do that, faculty needs to first create (or select from existing) a university information and create (or select from existing) a course information (Fig. 2). Once those are selected, faculty needs to import student information (from Learning Management System) and select the year and semester of that class and then can save the class in the home folder for it to be deployed later. Starting a class means uploading the class files in the cloud so that students can start logging in and also to let all the client devices know that there is a new class been started.

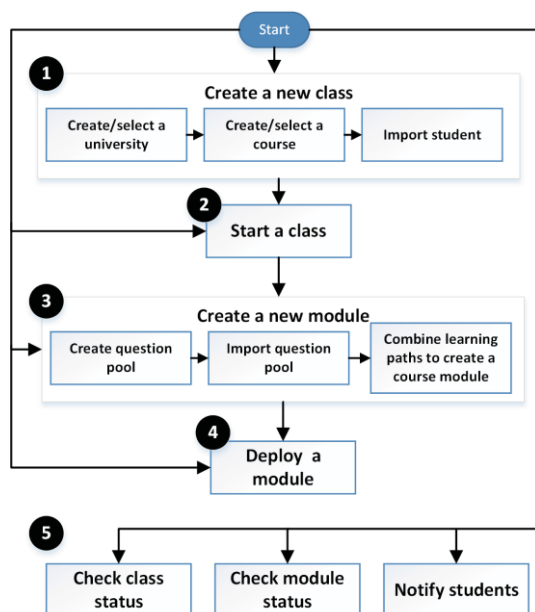


Fig. 7 Workflows in Dysgu.

Another important workflow for faculty is to create exercise modules for students to work on and then deploy the module to the class. As described earlier, A module is comprised of multiple smaller questions arranged in learning paths of different degree of difficulty. To formulate a module, the faculty first need to create the questions by utilizing the corresponding question generator. Fig. 8 shows a sample question generator for a specific type of problem-solving activity. Dysgu currently supports three different types of interactive problem-solving activities. However, Dysgu is developed in a way that it can be easily extended to support any new type of problem-solving activities as long as the designers of those activities follow some standards and interfaces. Using the question generator, faculty saves a pool of questions in a file, which is later used to design a module using the module builder interface (Fig. 9).

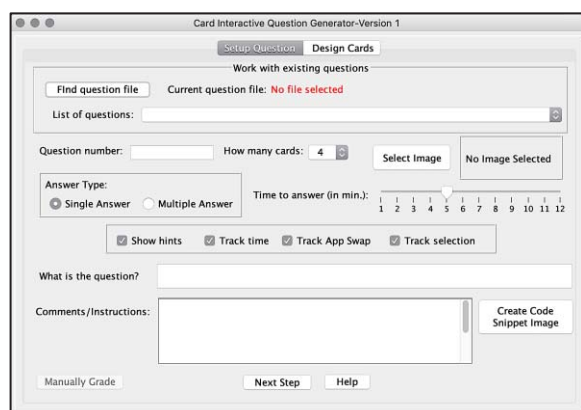


Fig. 8 Question generator.

The module builder (Fig. 9) helps faculty to formulate a problem-solving module, as described in previous sections. Once faculty import a question pool file, they are allowed to drag and drop the questions into the corresponding path. Problems in a path are provided to students as bottom-up and therefore the expectation is that faculty will put easier problems at the beginning of the paths and will make them

harder as they progress along that learning path to facilitate instructional scaffolding.

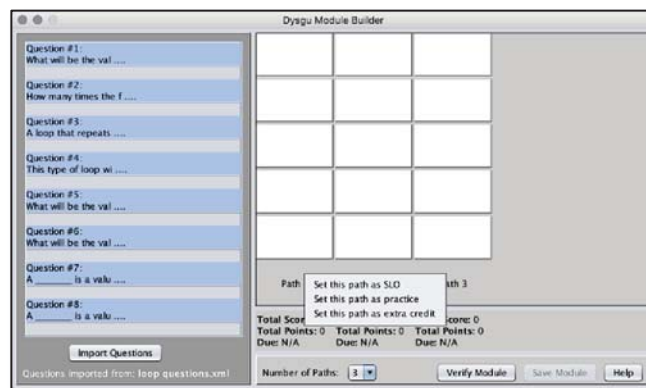


Fig. 9 Module maker interface.

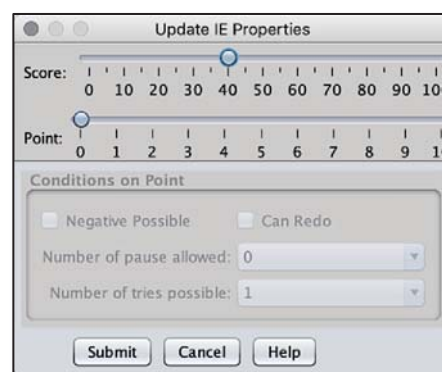


Fig. 10 Path properties.

Each path can have a deadline and faculty can either individually or across the modules assign a deadline by clicking on that path. Faculty needs to fine-tune each problem on a path by setting different properties of that problem (Fig. 10). If a point is assigned to a problem, the faculty needs to set different policies (such as whether the point can also be negative, whether students can redo a problem etc.).

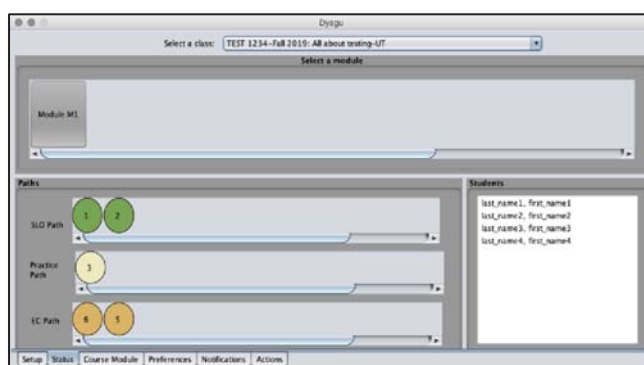


Fig.11 Class status interface.

As students start solving problems, faculty can monitor (Fig. 11) class progress on a module, on an individual question or even on a particular student. In this screen, when faculty clicks any of the module, the paths and activities in that module are shown. Then, when any of the activities are clicked, another dialog box (Fig. 12) shows details about that particular activity and a pie chart on the grades about how the class did on that activity. Therefore, faculty gets a glimpse of how the class are doing on a specific problem, which problems the students are solving successfully and where

they might be having problems. Additionally, faculty can click on any student in the class and another dialog box will open (Fig. 13) that lists grade statistics of that particular student. Faculty can analyze different aspect of the student's performance with regards to module, learning paths or activity.

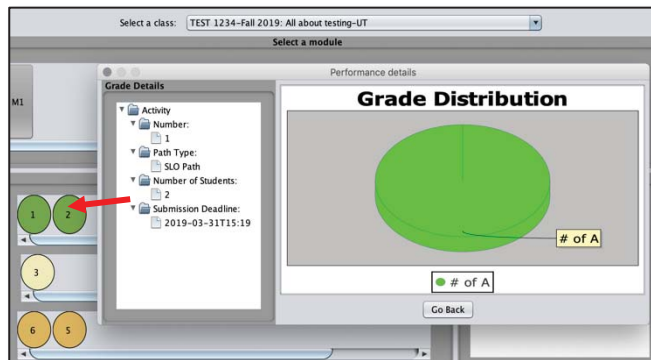


Fig. 12 Activity grade distribution.

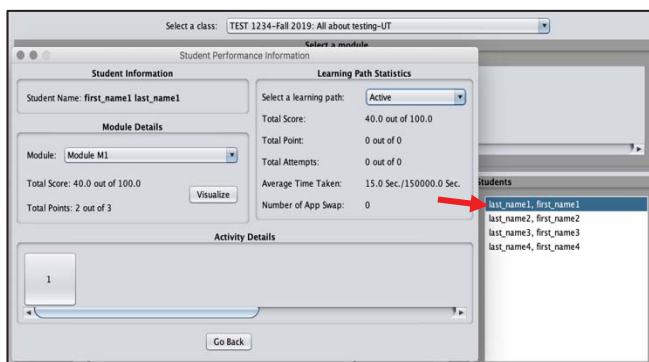


Fig. 13 Student performance detail.

On the student mobile devices, students first have to log in to a class by using their credentials. Once successfully logged in (Fig. 3), students can find current course modules, check on their status, setup schedule or other settings, see any existing or new text messages from faculty or can send a text message to the faculty.

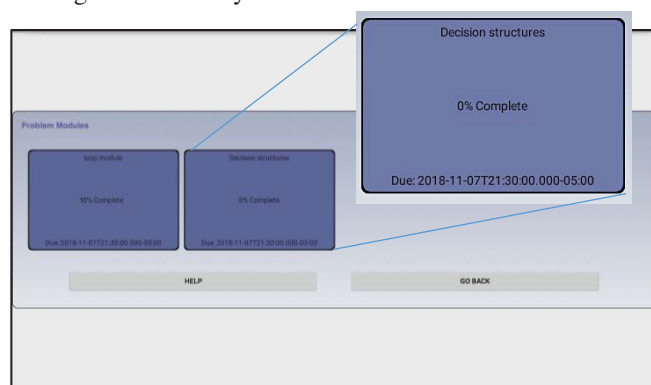


Fig. 14 Course module screen.

A student can check available problem modules and the deadline to finish a module (Fig. 14). Even if a module's due date is over, it will still be available in this list for the student to review the answer. When one of the modules is selected, another window opens (Fig. 15), that shows the learning paths in that module and individual problems in each path, which they can select to solve. If a problem is selected, a

corresponding interactive exercise activity app will run (Section III.A) that allows students to solve the problem.

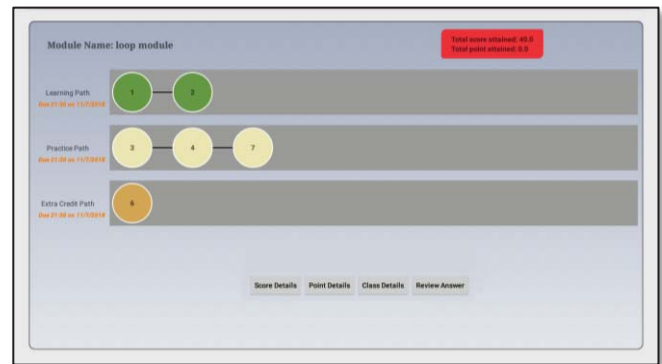


Fig. 15 Individual module screen.

A student can check their progress (Fig. 16) on different modules and can compare their own progress with the rest of the class. Dysgu shows the student score, their placement (module, path, problem specific) compared to the class, timing information about problem-solving activities, and points and badges the student earned. Students can also review the finished activity and compare their own performance with the rest of the class. Additionally, students are awarded badges depending on different conditions (fastest in the class to answer, highest score, etc.) and they can see what badges that they won and compare them with the rest of the class.

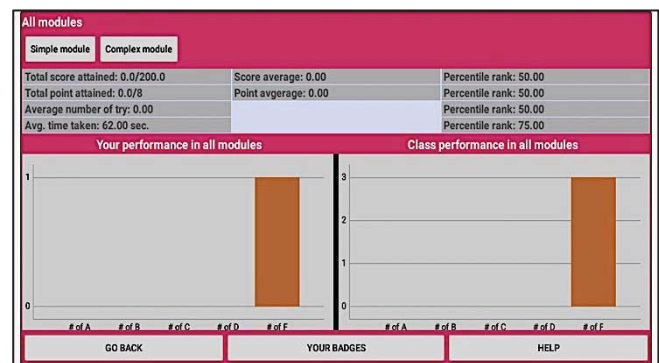


Fig. 16 Grade detail screen.



Fig. 17 View badge details.

Depending on the student's schedule (which students can set in the software), the app reminds students about pending activities and their deadlines or other encouraging or cautioning notifications (Fig. 18). Based on the privacy settings and student's information, the app recognizes the geophysical location and alerts the students accordingly. For instance, if the app recognizes that the student is in a restaurant while in a scheduled study time, it reminds the

student frequently about incomplete work; however, it might be silent when the student is at home or work. As students continue solving problems, they will earn badges depending on different criteria (such as time taken, highest in a module or in the class, most score or points, etc.) and they can not only show their own badges, but also can see other's in the class and compare their performance relative to the rest of the class. By providing such information about peers and by setting higher expectations for these types of activities (in terms of competition, problems with higher degree of difficulty, chance of gaining points, etc.) than traditional pen-and-paper activities, the expectation is to promote a self-fulfilling prophecy, which seems to work as shown in [7].



Fig. 18 Sample Dysgu notifications.

IV. SIMILAR SYSTEMS

A plenty of research have been performed to improve student's in-class learning and engagement, however, in comparison there is not much work done to improve traditional out-of-class student learning and engagement, explicitly, which incorporates mobile technology. A prototype mobile learning environment [8] is developed to reinforce student learning for out-of-class activities in a programming course. This system supports multiple-choice questions only and allows a student to answer a specific question per day, even if the student finishes it up right away. [9] have presented a collaborative online learning environment to link in and out of class activities and to allow student to collaborate using social networking and gamification components. In this system, all questions are narrative or discussions oriented and tied to a specific reading material set by the faculty. This system is browser-based, which lacks the flexibility and interactivity provided by mobile apps. Work presented by [10] combined both social networking and gamification to instruction and showed that it improves student's academic performance. De-Marcos's work has similar goal as the proposed study, however the delivery, mode of interaction, type of activity and nature of engagement are different in Dysgu. Additionally, some commercial systems such as Piazza [11], Prulu [12], Quizlet [13], and Socrative [14] that are similar in nature to Dysgu, however; they are different in design philosophy, scope,

activity type, pedagogical intentions, technique used, deployment platform or intended audience. The design goals of Dysgu addressed several critical dimensions necessary for creating an engaging learning platform. One of those goals were to have the ability to provide features that contributes to student learning, such as personalization through instructional scaffolding and frequent feedback. Secondly, to support the ability to deliver easily manageable and quality content for students, such as parameterized problems and problems with different degree of difficulties. Next, to have interactivity in such content so that students are more engaged in them, for instance, multiple screen problem-solving activity where students can traverse and see effects of their interactivity. Dysgu also provides adaptability by the way of sensing student's situational context (such as geo-location, personal schedule etc.) and provides contextual feedback and suggestions to acclimate to student's needs. Dysgu promotes motivating factors that brings students back and engage them to solve problems, such as game-centric currency that allows students to buy extra time or hints to solve a harder problem. Finally, Dysgu provides a dimension to compare student's own performance to the rest of the class, which allow students to feel an integral part of the big class and not alone solving the problems. Flipped classes [15] became popular recently because of its perceived benefits in student learning and engagement. However, there are concerns [16]-[17] related to flipping classes, especially that it might not work for students who come from socially disadvantaged background. Dysgu can address those concerns by its nature of pervasive access, interactive nature and motivation-oriented environment to augment flipped classes in order for it to become more productive for a wide variety of students.

V. EVALUATION AND CURRENT STATUS

The current version of the software is developed using Java and Android. Although the software hasn't been used in a class yet, a small group of undergraduate students helped to pilot the system and provided a lot of useful feedback that was utilized to update the design and which showed that the system is useful to them. A sample of the comments from the students are as follow:

"I can start working on the activity any time and see what my classmates are doing and compare my progress"

"Dysgu is intuitive to use. Although I want it to look more professional, it does the job and easy to use"

The pilot study also provided a lot of inputs about the design of the user interface and by combining several mobile learning requirement catalogs [18]-[19], Dysgu was designed to be more responsive, attractive and motivating.

To examine the quality and maintainability of the code, we use a static code analysis tool named CodeMR [20], which analyzes the code structure, measures different quality matrices of the code, allows visualizing the design, and reports design flaws. CodeMR provides a set of software quality metrics, suitable to cover the most important aspects of structural quality. It is the belief of the developer that, any

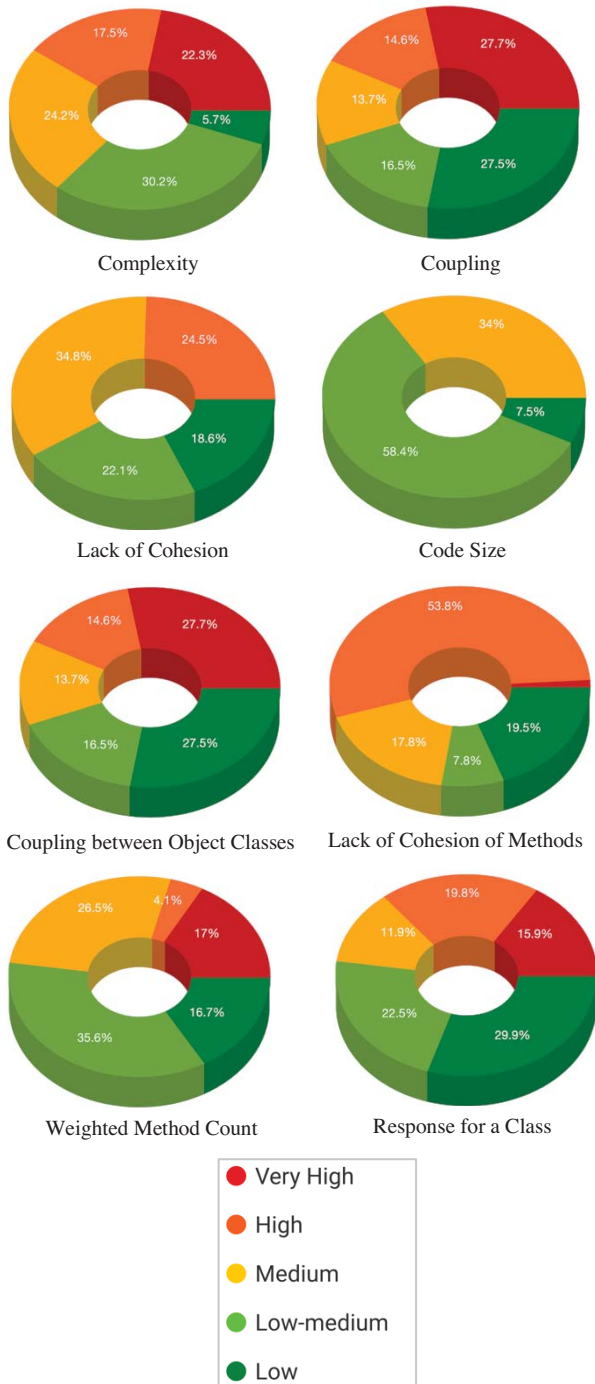


Fig. 19. Distribution of software quality attributes.

software designed for educational purposes not only have to be effective in student learning and engagement but also should have good software quality properties to make it easily manageable, maintainable, testable and extensible. Fig. 19 shows the object-oriented software quality metrics [21] (% of the code) generated by CodeMR using static code analysis. These measures are used to determine the code's structural complexity. Studies [22]-[23] show a correlation between a program's quality metrics and code's maintainability and extensibility. In that regard, it is evident from Fig. 19 that the values for Dysgu falls within the expectable threshold [24] for software to be easy to maintain and to extend. However, as shown in Table I, some of the metrics' value for Dysgu has

some room for improvement and will be investigated more in the future to achieve that.

Once Dysgu is used in classes, the project will execute a Mixed Method Convergent design [25] that will incorporate both quantitative and qualitative approaches and use results from both to address evaluation questions. Student performance and course data gathered with the new intervention in the target courses will be compared with the prior offerings of the courses taught in a traditional way. Pre-post survey will include questions on course modules and learning environment software, student confidence and satisfaction and their perception of learning with technology.

TABLE I. QUALITY METRICES RANGE FOR DYSGU.

Matrix	Goal	Status
Complexity	Low to Medium	58.3% is within the goal
Coupling	Low to Medium	57.3% is within the goal
Lack of Cohesion	Medium to Very High	59.3% is within the goal
Code Size	Low	65.9% is within the goal
Coupling Between Objects	Low to Medium	57.7% is within the goal
LOC Methods	Medium to Very High	72.7% is within the goal
Weighted Method Count	Low to Medium	78.8% is within the goal
Response for a Class	Low to Medium	64.3% is within the goal

VI. CONCLUSION

This paper presents a new mobile-based out-of-class learning environment called Dysgu, where students actively participate to solve interactive problems with scaffolding, personalization and engage in self-reflections and game-centric interactions. Dysgu utilizes the transformative power of mobile technology for out-of-class activities, student engagement in active learning outside the classroom, and the possibility of providing early intervention.

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